Lecture plan · for the lecturer

IoT protocols: MQTT, CoAP and LoRa

IoT · University · Lecture 3

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Explain publish/subscribe.
- Choose MQTT, CoAP and a QoS.
- Trade range against rate.

Activity: A broker in the room

1. Warm-up (5 min): Give the farm brief and ask teams to choose a network in two minutes.
2. Set up (10 min): Run a broker on a laptop and share its address.
3. Publish (20 min): Teams publish readings at QoS 0 and 1 and subscribe to another team.
4. Choose (20 min): Teams pick a link and protocol for one brief and present the trade-offs.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- When would you pay for QoS 2?
- What breaks if the broker fails?
- Which brief was hardest, and why?

Exit ticket

- Choose a link and QoS for a smoke alarm in a flat, and justify it.

Try it in your next lecture
Run a free MQTT broker on a laptop; each team publishes sensor readings to its own topic and subscribes to another team's.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **40 hectares, no power, a few bytes every 15 minutes. Which link?**

◯ Bluetooth Low Energy ◯ LoRa ◯ Wi-Fi

2 **Count the messages.**

One publish at QoS 1 needs _____ packets between client and broker; at QoS 2 it needs _____.

Word bank: 3 · 4 · 2 · 1

3 **Which situations favour CoAP over MQTT? Choose all that apply.**

☐ UDP is required to avoid TCP overhead

☐ A gateway queries constrained nodes for readings on demand

☐ Devices must expose resources like a tiny web server

☐ Many dashboards subscribe to the same stream

4 **LoRa's long range comes at the cost of a low data rate.**

◯ True ◯ False

5 **Match each brief to the most suitable link.**

Wearable to phone, 2 m	• • • •	Zigbee or Thread mesh
Camera streaming video indoors	• • • •	LoRaWAN
Hundreds of lights in one building	• • • •	Bluetooth Low Energy
Soil sensors over several km	• • • •	Wi-Fi

Answers · for the lecturer

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1 **LoRa**

LoRa reaches kilometres on little energy, and the data need is tiny.

2 **One publish at QoS 1 needs 2 packets between client and broker; at QoS 2 it needs 4.**

QoS 1: PUBLISH + PUBACK. QoS 2: PUBLISH, PUBREC, PUBREL, PUBCOMP.

3 **A gateway queries constrained nodes for readings on demand; Devices must expose resources like a tiny web server; UDP is required to avoid TCP overhead**

Request/response over UDP fits queried resources; many subscribers to one stream is MQTT's strength.

4 **True**

Spreading the signal over time improves sensitivity but cuts throughput to a few kbit/s.

5 **Wearable to phone, 2 m — Bluetooth Low Energy**
Camera streaming video indoors — Wi-Fi
Hundreds of lights in one building — Zigbee or Thread mesh
Soil sensors over several km — LoRaWAN

Distance, data rate and power together pick the link.

For you to think about

How do you teach networking in your IoT course now? Could a design brief replace a protocol lecture?